Electrical and Electronics Engineering and Robotics

ELEC345 High Speed Communications

Double stub matching

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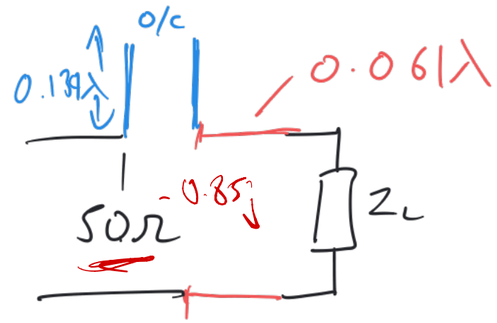
Remember - Single stub Match.

$Z_0 = 50\Omega$

$Z_L = 30 + 25j$

$z = 0.6 + 0.5j$



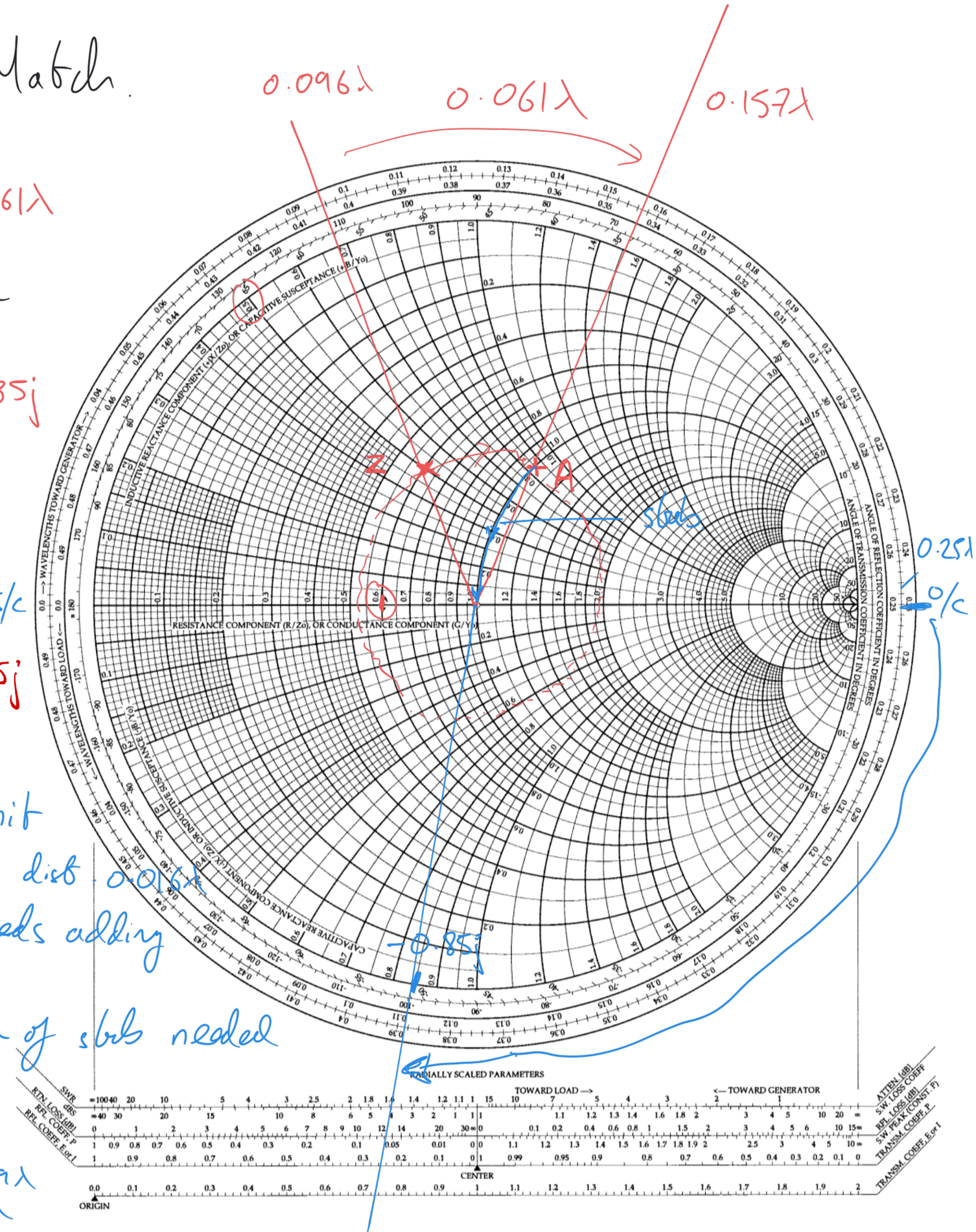
$A = 1 + 0.85j$

Stub must equal $-0.85j$

o/c stub $0.389\lambda - 0.25\lambda$

$= 0.139\lambda$

s/c $= -0.85j$



① Rotate twd gen round VSWR circle until you hit matching circle. Read off dist. 0.016λ

② Look at how much j needs adding or removing. $(-0.85j)$

③ Choose type & find length of stub needed to give this o/c. twd gen from $0.25\lambda \rightarrow 0.389\lambda$
$= 0.139\lambda$

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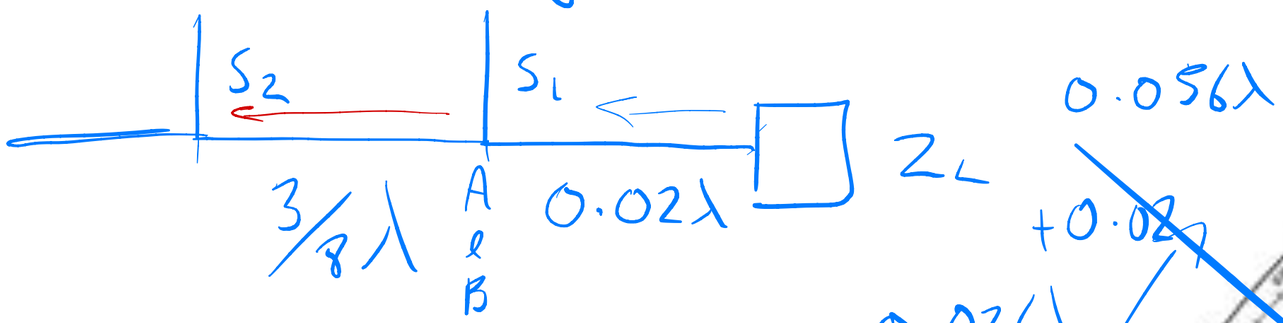
$Z_L = 20 + 10j$ Normalize $(\div 50) \Rightarrow 0.4 + 0.2j$



0.056λ

$+0.02\lambda$

0.036λ

To A is $0.3j \rightarrow -0.15j = -0.45j$

To B is $0.3j \rightarrow -1.9j = -2.2j$

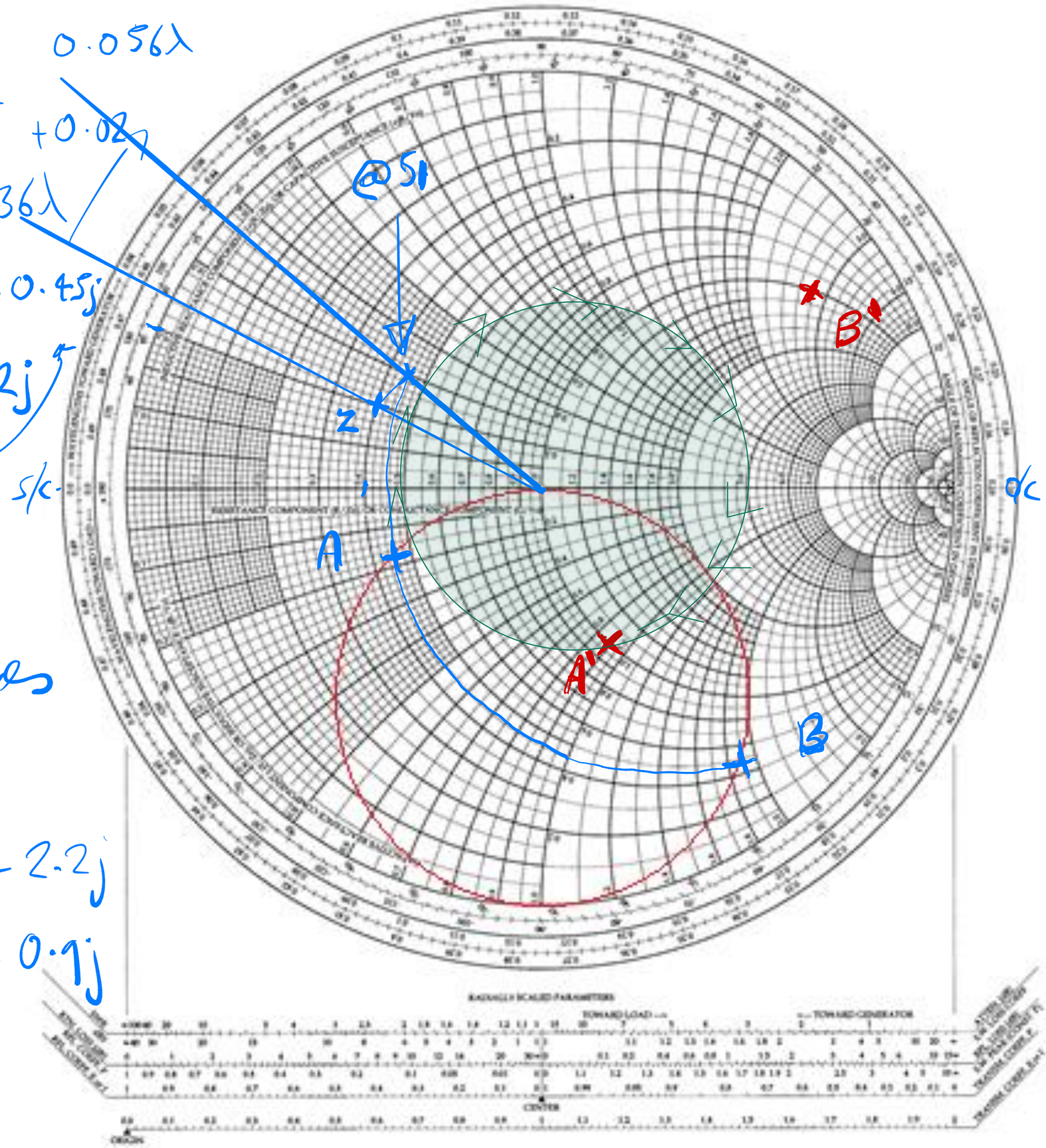
use o/c stub for either

$A' = 1 - 0.9j$ $B' = 1 + 2.9j$

Lastly 2nd stub removes $\pm j$ from A' & B'

$1 + 2.2j$ so o/c stub for $-2.2j$

$1 - 0.9$ so use s/c stub $+0.9j$



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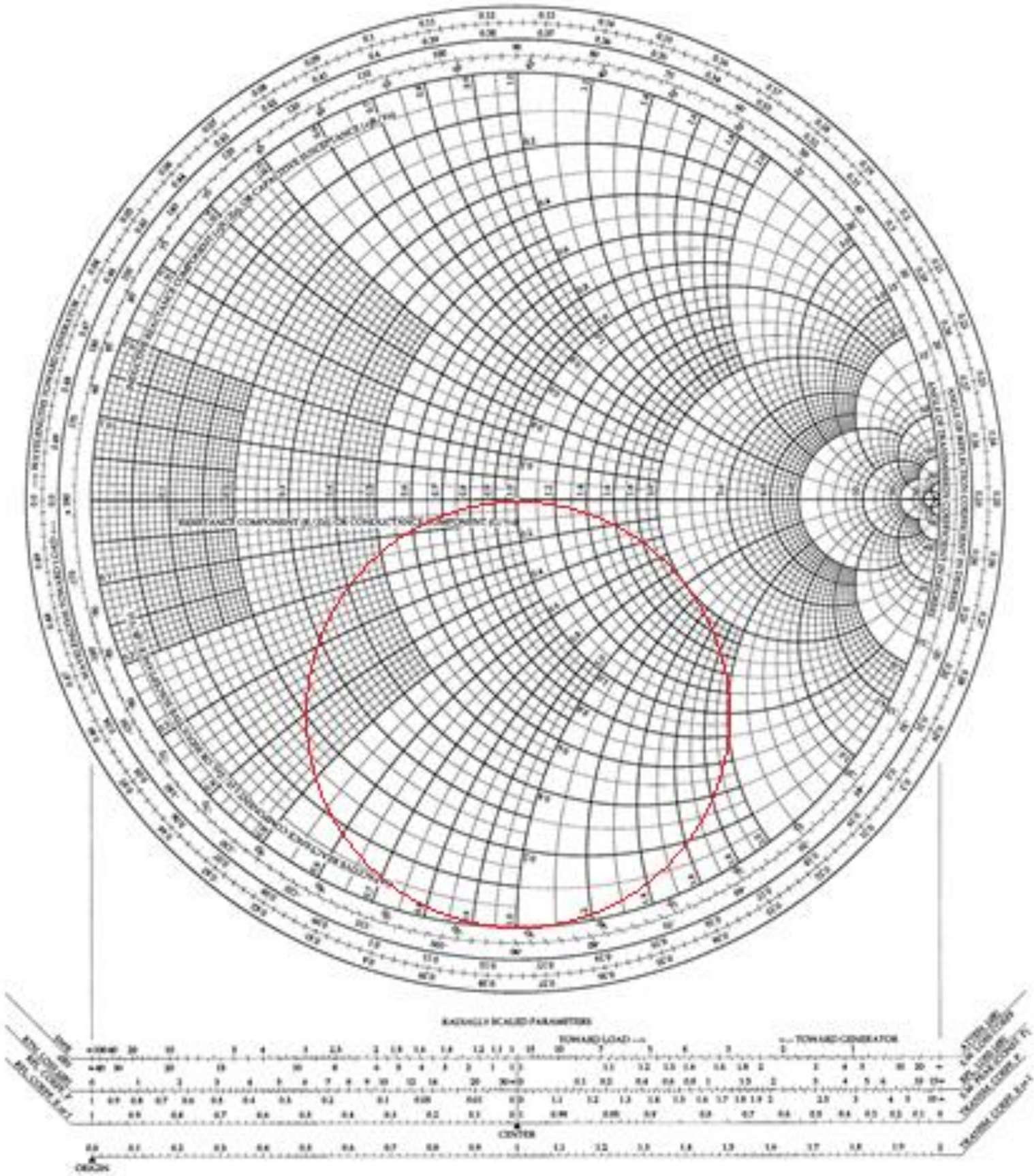

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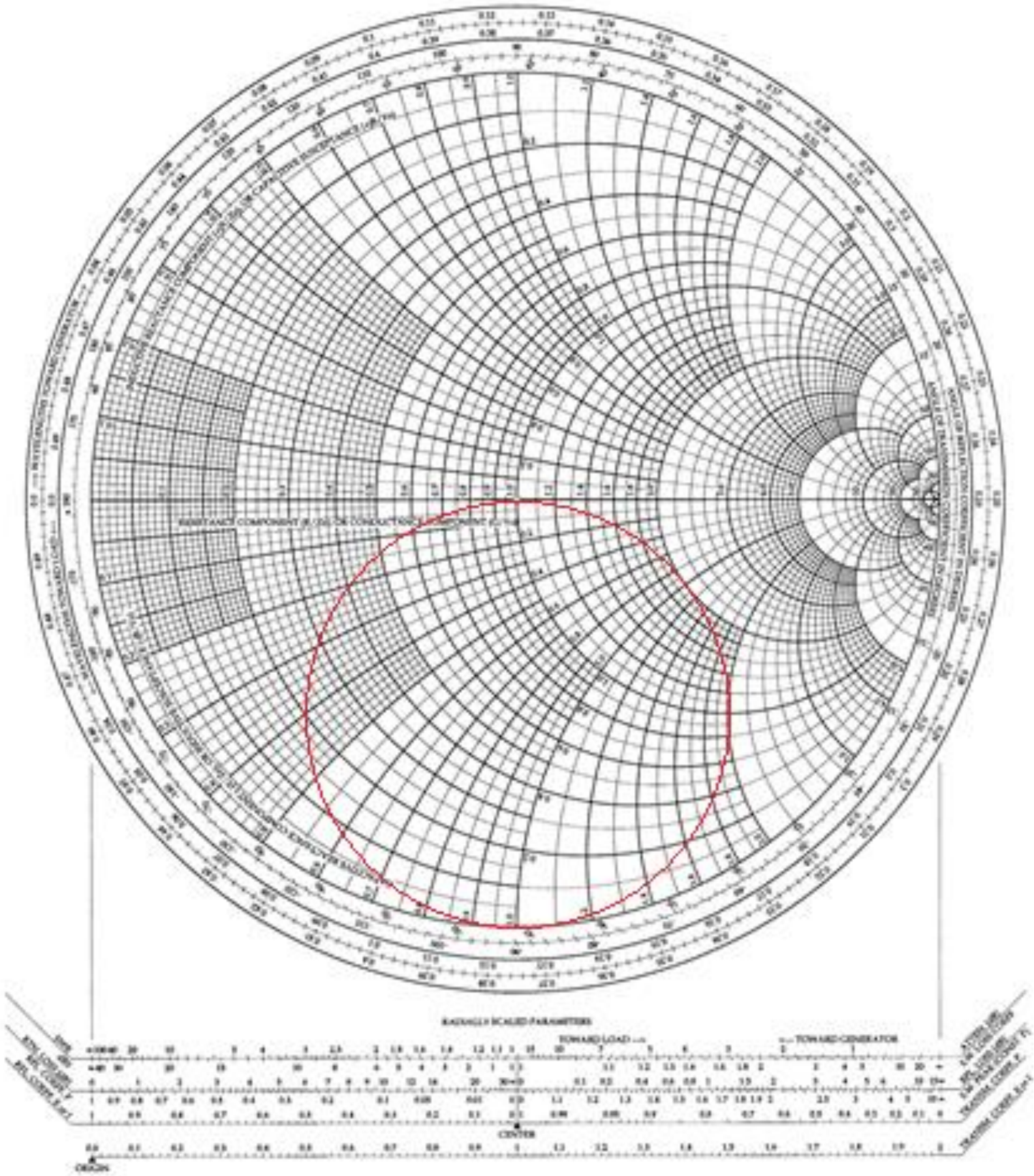
Parallel Double stub match steps

1. Plot load. $Y_L \times 50$ $Z_L \div 50$ then transpose to Y_L
2. Rotate to the first stub S_1 often 0.02λ but can be a choice
3. Use stub 1 to add or subtract j onto $3/8\lambda$ guide circle
4. Calculate length of stub for this j
5. Rotate $3/8\lambda$ to stub 2 and onto matching circle.
6. Use stub 2 to add or subtract j to centre
7. Calculate length of stub 2 for required j.

Remember to transpose s/c & o/c for parallel matching



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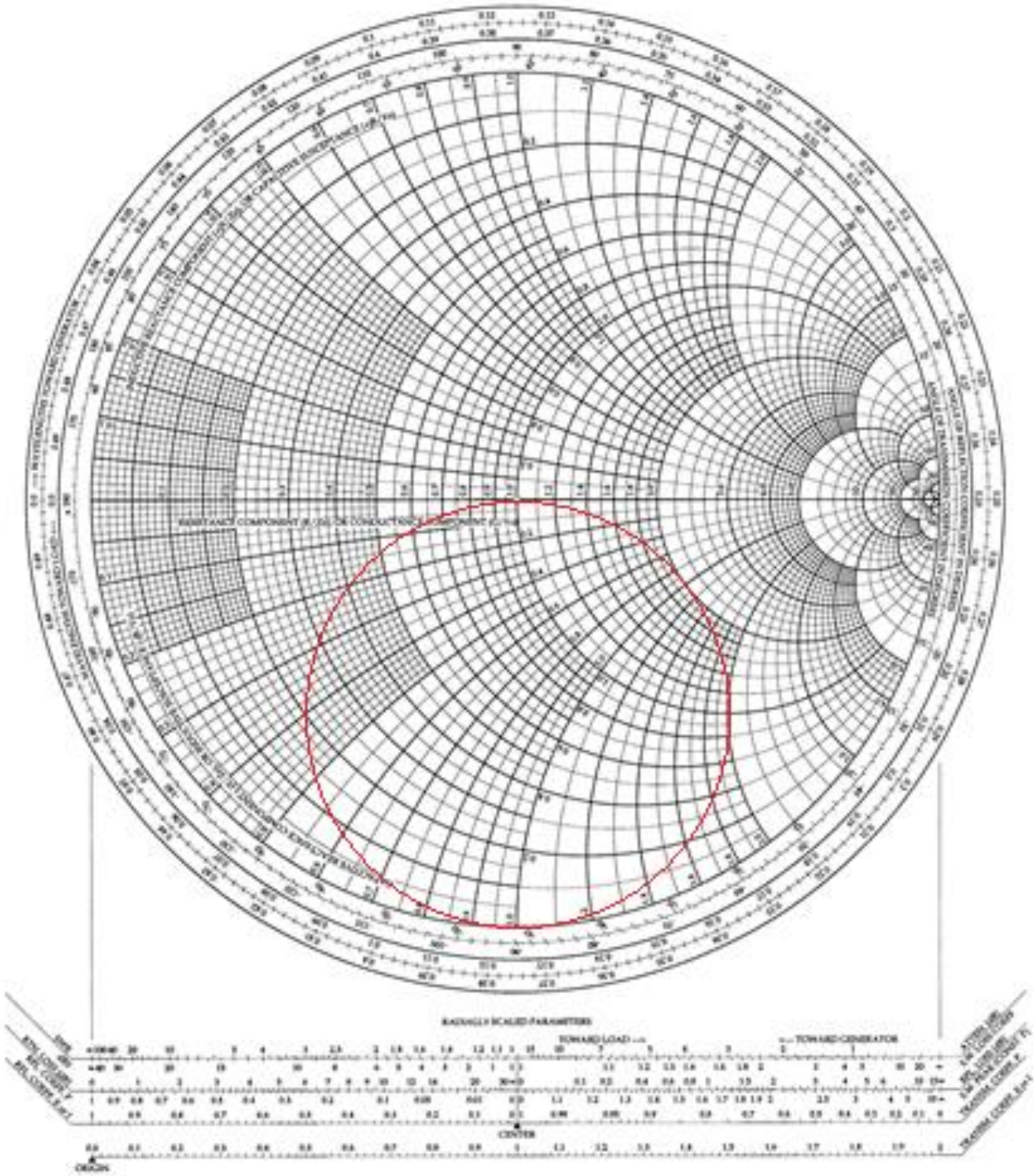

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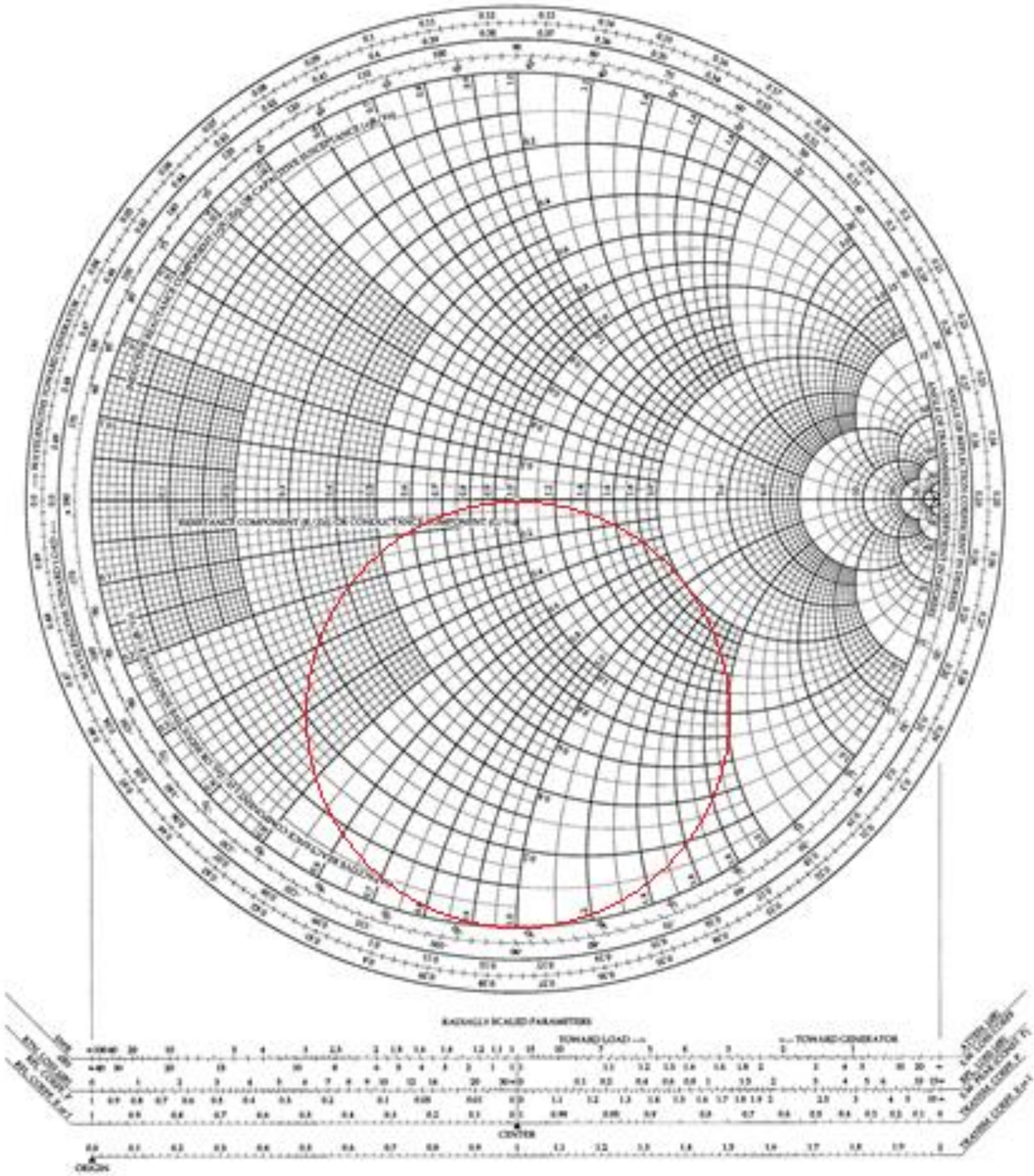

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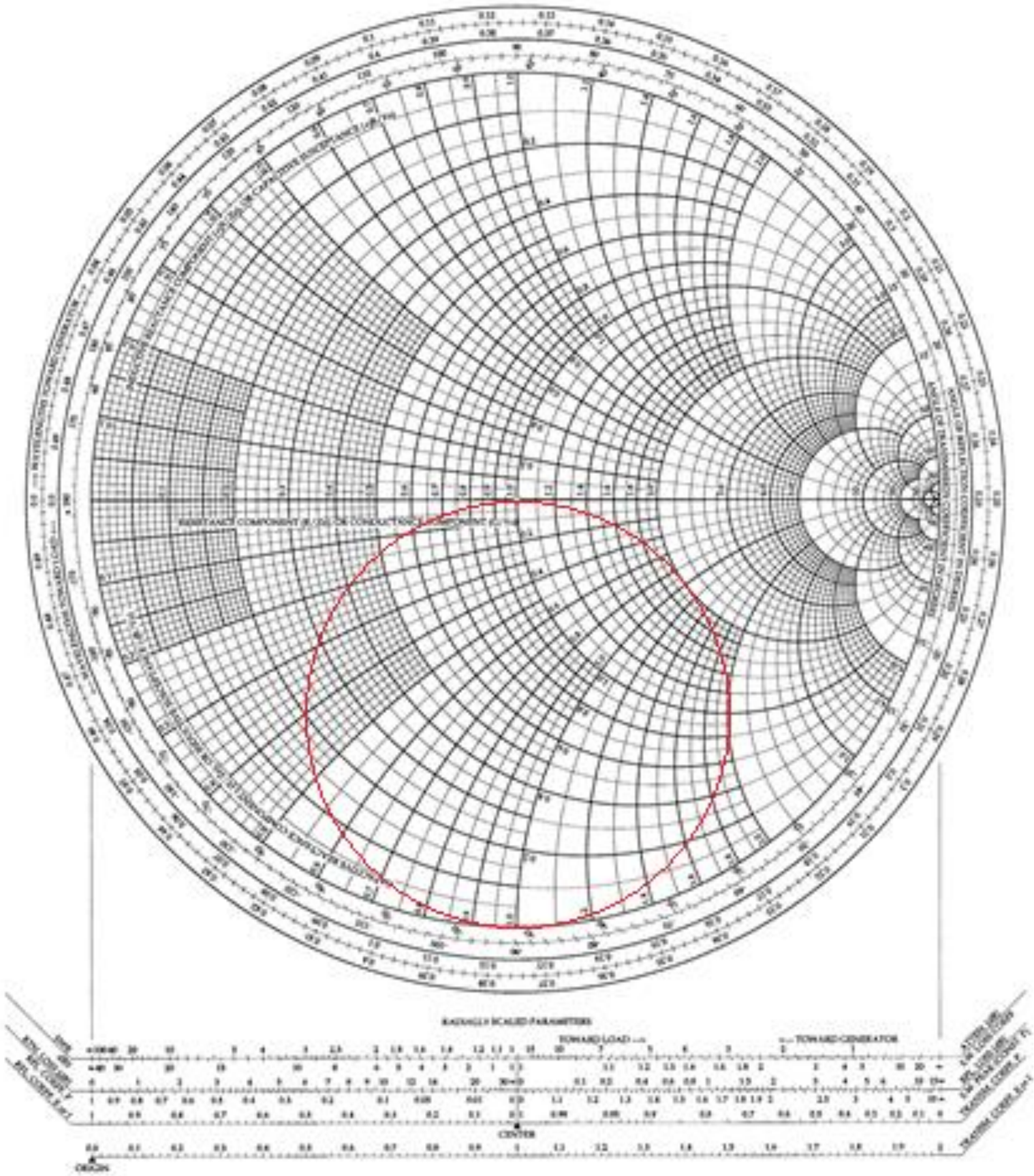

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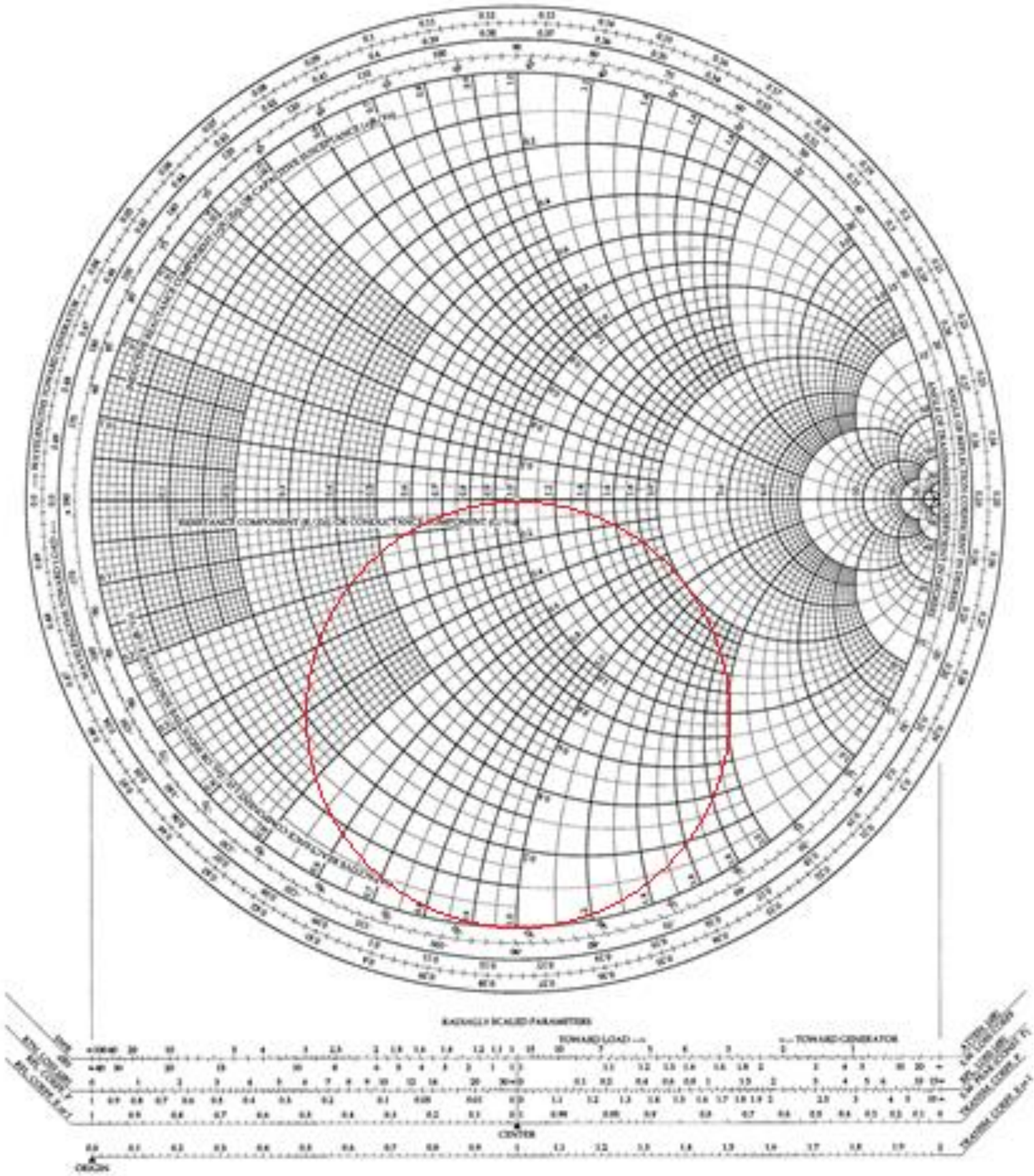

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